

Balanced coarse sums on anchored split frames

A conditional integer-multiplication saving of $5.1547855889988 \times 10^{-5}$

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With substantial OpenAI Codex assistance

October 9, 2026

Abstract

We give a checked finite composition of the balanced coarse-column sums of PR69 with the anchored split-pair graph of PR71. A cost-ordered dependency basis is used when reclaiming scratch carriers. At dimensions 23 and 25 the result has 27,338 and 35,939 auxiliary roles. Under the inherited analytic, all-size framed-word, residual-compiler, routing, exact-recovery and fixed finite-alphabet multitape interfaces, it yields $T(n) = O(n(\log n)^{1-\kappa})$, where $\kappa = 12886963972497/250000000000000000$. This is a 0.2590891819% improvement over the pinned PR71 witness. The complete old child list is excluded at the new bit saving, while the new list passes exact rational bounds. The finite tests and this conditional argument are not a formalization or independent expert review of the full multiplication theorem.

1 Construction and inherited interfaces

We retain the original-envelope geometry, simultaneous rational $I + J$ basis, two-stage data network and complex branch of the cited construction chain. Address geometry is rational; the scalar payloads in the bit branch are over $\mathbb{F}_2$. The graph is Avi Eisenberg’s interval-strip and core-aware pair graph [4], with the split operation of Rohan Garg [2] and the anchored choices of Chafik Boukhalfa [8].

At each common point, start with the inherited alternating point order and move the lowest intact global pair to the beginning, preserving the other points’ relative order. At the first three recursive levels use split vector $[1, 1, 2]$: split the first pair at the first two levels and the middle group at the third when it is a pair. Splitting is disabled when at most three points remain. The groups have size one or two, partition the current point set, and strictly reduce its size. Later levels retain ordinary pairing. Region order is **cover-core** at $h = 23$ and **reverse-node** at $h = 25$, as in PR71.

The new composition uses eumemic’s balanced coarse-column identity [7] inside this split graph. For disjoint groups G_i, G_j , set

$$c_{ij} = \sum_{b \in G_j} \left(\sum_{a \in G_i} e(a, b) \right). \quad (1)$$

This equals the previous flat sum, with every edge appearing once. The two column sums at a pair-pair interaction are interned by their actual supports and can be shared with existing strip computations. Equation (1) also applies when either group is a singleton. No producer output or data pair is removed. Every realized support, scalar output and frame inclusion is checked again.

The chosen region compiler is descended from eumemic’s reversible compiler [1], with Chafik Boukhalfa’s reclaimed carriers [3], Rohan Arun’s profile-cost oracle [5], and Dominik Scholz’s

compatible pending controls [6]. These are inherited contributions. Our finite witness specifies their composition with (1), and orders eligible live and retired basis candidates by their actual profile cost before selecting a clearing dependency. This selection differs from considering retired candidates only in descending frame rank. No optimal matching or global search theorem is used.

2 Valid clearing with a different dependency basis

Write F_s for a carrier's current frame and E for the frame of the current region. Retired candidates satisfy $F_s \subseteq E$. A pending live control with scheduled next-use frame G_s is eligible only if

$$F_s \subseteq E \subseteq G_s. \quad (2)$$

The temporary binary elimination basis contains current region anchors, eligible live controls, and independent eligible retired candidates. If a candidate's clean signal is dependent, elimination supplies a literal expression $v_s = \sum_{a \in A} v_a$, with $s \notin A$.

Proposition 1. *Changing the order of eligible basis candidates and selecting any resulting dependency preserves the reversible dirty-scratch interface, provided every selected operand raise is recorded and charged.*

Proof. Searching and comparing dependencies changes no physical value or frame. For the selected expression, emit the two-carrier XORs $s \leftarrow s + a$, for $a \in A$, at frame E . Each is invertible, raises both operands through containing frames, and leaves the control's scalar value unchanged. The clean signal on s becomes zero; its arbitrary dirty part need not be zero. A live control retains its future validity by (2). Removing s from the retired set and reusing it therefore introduces neither a free copy nor a zero-initialized auxiliary assumption. All frame incidences appear in the actual serialized word. $\square$

For discovery only, let $H(F, G)$ denote the sum of $t \log t$ over the actual fixed-basis blocks of the transition $F \rightarrow G$, evaluated on a fixed integer grid using rational logarithm enclosures. A live control is priced by

$$H(F_s, G_s) - H(F_s, E) - H(E, G_s), \quad (3)$$

and a retired carrier is priced with the full terminal frame in place of G_s . Eligible controls and retired candidates are sorted by this cost; the complete stored dependency is scored with all operands included. The earliest candidate order is used only to construct a basis and its expressions. Every dependent candidate is considered before selecting the lowest scored expression. Equation (3) predicts local fragmentation and does not assert the cost of unknown later uses. Only the final complete profile is an acceptance criterion.

Let L be the resulting invertible mixer, V the source injection and J the terminal scatter. The complete fresh-source check establishes $JLV = I$. For source x , target y and arbitrary scratch d , the chronological word

$$L, J, L^{-1}, V, L, J, L^{-1}, V \quad (4)$$

adds $JLd + JL(d + Vx) = x$ to the target and restores the scratch over $\mathbb{F}_2$. The reversed transposed word gives the other orientation. The literal terminal ports, including all retained totals, are checked independently; net cancellation does not excuse uncharged extra scatter gates.

3 Complete physical accounting

At $h \in \{23, 25\}$, let $v_h = \binom{h}{3}$ and let R_h be the auxiliary role count. Each local profile includes every actual operand promotion, copied retained center, ordinary output growth and final cleanup. Its rank mass satisfies

$$\sum_t tn_{h,t} = hR_h + h(h-1). \quad (5)$$

Both the event log and transitions reconstructed from the literal XOR word give the same profile input. The unchanged $I + J$ profiler computes its ordered contiguous blocks with the inherited bounded-minor and three-prime CRT checks. There are no modular disagreements.

Quantity	$h = 23$	$h = 25$
Sources v_h	1,771	2,300
Auxiliary roles R_h	27,338	35,939
Mixer XORs	161,829	226,834
Clearing XORs (included)	3,609	5,455
Wrapped word length $4 L + 14v_h$	672,110	939,536
Input, target and dirty basis columns	30,880	40,539
Local rank mass	629,280	899,075

Put $m = 23 \cdot 25 = 575$ and $N = v_{23}v_{25} = 4,073,300$. Replicate the h23 profile 2,300 times and the h25 profile 1,771 times. Retain the full inherited data profile, including all N data pairs and the previously recovered exceptional pairs. Then

$$\begin{aligned} W &= 2N + 2300R_{23} + 1771R_{25} = 134,671,969, \\ L_0 &= 2300 \cdot 23 \cdot 22 + 1771 \cdot 25 \cdot 24 = 2,226,400, \\ s &= mW - N + L_0 = 77,434,535,275, \\ mW - s &= 1,846,900. \end{aligned} \quad (6)$$

The global profile includes two data profiles $9[1] + [21] + [17] + [481]$ per data pair, one paid rank-one endpoint copy per pair, both local profiles, and all exterior and source-growth classes. Every child width is positive and less than 575; the maximum is 529. The complete integer multiplicities are stored in `selected/certificate.json`; none is inferred by scaling a previous exponent or by role count alone.

4 Exact exponent and conditional assembly

For saving $a > 0$, the characteristic of the complete bit profile is

$$F(a) = \frac{1}{mW} \sum_t n_t t \exp\left(a \log \frac{m}{t}\right). \quad (7)$$

The selected witness is

$$a = \frac{51550513208569}{10^{18}}. \quad (8)$$

All logarithms are enclosed by a rational atanh series with explicit remainder. For $0 \leq u \leq v < 1$, use the degree-eight Taylor lower bound at u , and the upper bound

$$\sum_{j=0}^8 \frac{v^j}{j!} + \frac{v^9}{9!(1-v/10)} \quad (9)$$

at v . The geometric remainder follows from the ratio of successive omitted terms. Directed rounding widens the intervals. The exact accepted upper moment is below one; the lower moment at $a + 10^{-18}$ is above one. A second evaluation uses powers of $3/2$ for logarithm range reduction, 60 atanh terms and a degree-twelve exponential bound, and confirms both signs.

The complete PR71 profile, with its own width, has a lower moment above one at (8). Thus the new finite profile gives a strict structural improvement. The comparison is to PR71 commit `1bef94fd40a746452548c84a4a8f8834670a3113`.

Theorem 1 (Conditional on the inherited interfaces). *Assume the base integer-multiplication theorem and the all-size ordered residual compiler, framed-word transfer, fixed finite-alphabet multitape layout, analytic and semantic reductions, routing, eligible-prime setup and exact-recovery interfaces used by the cited construction chain. Then*

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{12886963972497}{250000000000000000}.$$

Proof. Use (4)–(8) as the new fixed bit-network interface in the inherited balanced assembly. The complex saving remains $717/10^7$, $\beta = 1/20$, and the semantic guard remains one. Set $\eta = 10^{-12}$, $q = a(1 - 2\eta)$, $\epsilon = (1 - \eta)/(1 + q)$. The controlling margin is $G = \epsilon q$. The certificate recomputes all 47 strict constraints and seven margins and verifies $\kappa < G < \kappa + 10^{-18}$. The bit wire bound remains 28 bits and the bit halving degree remains nine: $2 \cdot 529^9 < 575^9$, whereas $2 \cdot 529^8 \geq 575^8$. The unchanged complex branch retains its literal scalar charge and halving degree. The row-stock and semantic bounds are consequently valid with the updated W ; their exact inequalities and sufficient arithmetic cutoffs are reevaluated. The positive final slack permits the inherited logarithmic absorption. Arithmetic cutoffs alone are not a complete operational threshold. $\square$

5 Reproduction and limits

Run `make -C research/balanced-split-reclamation verify` to regenerate both physical words byte-for-byte, replay every input and dirty basis column in both orientations, rebuild the actual profiles and exact certificate, and run failure-mode controls. Run the repository’s `make -j1 verify` for its broader retained checks. The twelve focused tests include malformed indices, aliased sources, understated width, omitted paid children, wrong terminal ports, canceling uncharged scatter additions and adjacent exponent grid rejection. Source hashes are frozen explicitly; verification does not refresh them.

These checks verify fixed finite objects and inequalities. They do not execute the complete asymptotic multiplication machine, establish global optimality, formalize the inherited interfaces, or demonstrate a practical speedup. This work used substantial OpenAI Codex assistance; its additional checks are not an independent human referee report. Contributor code, licenses and notices remain preserved. The new result is the concrete composition and its checked improvement, not authorship of the inherited ingredients.

References

- [1] eumemic. Joint reversible frame compiler, PR57. <https://github.com/CrocSwap/integer-mult-bounds/pull/57>.
- [2] Rohan Garg. Split-pair recursion, PR59. <https://github.com/CrocSwap/integer-mult-bounds/pull/59>.

- [3] Chafik Boukhalfa. Ranked slot reclamation, PR60. <https://github.com/CrocSwap/integer-mult-bounds/pull/60>.
- [4] Avi Eisenberg. Interval strips and core-aware pair assembly, PR62. <https://github.com/CrocSwap/integer-mult-bounds/pull/62>.
- [5] Rohan Arun. Cost-aware retired-slot selection, PR67, commit b3745601e947. <https://github.com/CrocSwap/integer-mult-bounds/pull/67>.
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- [7] eumemic. Balanced coarse columns, PR69, commit 91aa1f17e6e3. <https://github.com/CrocSwap/integer-mult-bounds/pull/69>.
- [8] Chafik Boukhalfa. Anchored split frames and next-use scoring, PR71, commit 1bef94fd40a7. <https://github.com/CrocSwap/integer-mult-bounds/pull/71>.